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AUTOMATION AND EDUCATION: I. General Aspects

• JAMES D. FINN

This is the first in a series of three articles that will discuss one of the most important and timely problems that educators face today—the effects of automation upon education. James D. Finn is professor of education at the University of Southern California and educational consultant for "Teaching Tools" and "Film World" magazines.

IN THE year 1869, Thomas Henry Huxley addressed the Liverpool Philomathic Society at a dinner meeting. His subject was scientific education. During the course of his speech, he said, "As industry attains higher stages of its development, as its processes become more complicated and refined, and competition more keen, the sciences are dragged in, one by one, to take their share in the fray; and he who can best avail himself of their help is the man who will come out uppermost in that struggle for existence. . . ." (21:114)

Huxley was a mighty prophet in many ways, but never was his crystal ball more clear than that night slightly less than one hundred years ago. Over the century since he spoke, the sciences and their offspring technologies have, indeed, one-by-one and two-by-two, entered the arena of industry (and of military science) until practitioners of the most unlikely sciences can be found busy on Madison Avenue, in the General Motors Research Institute, or the Pentagon Building.

Somewhere along the line, the First Industrial Revolution—which was the product of inventors—became the Second Industrial Revolution—which is the product of scientists and technicians. We are told on all sides by competent observers that we are just beginning to reap the harvest of the Second Industrial Revolution; that we have, in fact, just entered the era; that atomic power, rocket engines, and automation, for example, are in their technological infancies.

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Of all of these phenomena which have resulted from the application of science to industrial and military problems, automation is, or should be, of greatest interest to the professional educator. The problems and promises inherent in automation are directly related to education. Educators must re-examine science and technology in the second half of the Twentieth Century, keeping in mind Huxley's phrase that ". . . he who can best avail himself of their help is the man who will come out uppermost in that struggle for existence. . . ."

What is needed in 1957 and for many years to come is (a) a thorough assessment of the nature of the new technological world and its implications for education; (b) a revision of educational organization, procedures, and content, which will close the gap between the rocket-like progress of technology and the snail's pace of education; and (c) the application of useful concepts and processes from the new technology to the educational enterprise itself in order to close this ever widening gap.

Such a procedure will be a serious and difficult matter. It is not here suggested that educators take up a new fad. If automation and its related concepts are grasped by eager curriculum workers, school administrators, and professors in teacher education institutions, at the level of understanding and performance that some of the recent fads in education have been handled, we shall be worse off than before. We emerged from the "group dynamics" fad witnessing hundreds of people at various levels in education talking about a process they didn't understand and, fooled by their acquisition of the vocabulary, attempting to do things with group processes and with the results of group dynamics research that couldn't be done—as well as not doing other important things that should have been done.

We are still gripped today by a "3 R's" fad and a "gifted" fad which, presumably, will be dropped at the introduction of some other new idea that will make "experts" of its first practitioners long before they have an inkling of what they are either doing or talking about. Automation cannot be treated this way. To repeat, it is a serious and difficult business and offers no haven for the educator looking for something with which to impress the Teachers' Institute next fall.

Inasmuch as the audio-visual approach to communication is, strictly speaking, an attempt to solve a difficult human problem with technology, and, inasmuch as ideas have recently been

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advanced in several quarters that audio-visual technology might well become the means by which the process of education itself might become "automated,"[1] it follows that students of audio-visual communication must lead in the examination of the relationships of automation and education. The writer proposes to do this under three general headings, this paper covering the first. The three headings are: (a) General Aspects, (b) Automatizing the Classroom, and (c) Built-in Dangers. Future papers will deal with the other two topics.

In considering the topic of automation and education from the audio-visual point of view, there is a temptation to go directly to the central question continually raised by Dr. Eurich; namely, Can we, with audio-visual materials and devices such as film or TV, extend the range of the master teacher and replace the mediocre or non-existent teacher? However, because of the complex nature of the automation concept and because it will have other effects on education which, in turn, will affect the audio-visual movement, it is necessary to begin the study by analyzing the more general relationships of automation with education. This background study is the theme of this article.

The idea that the Second Industrial Revolution and automation, one of its main products, will have a profound effect on education is not exactly new. Statements abound to this effect, including one of the author's (14). However, the existence of ideas and their immediate or even consequent effect on educational practice is very difficult to establish, fads excepted. There is, as is well-known, generally thought to be a fifty-year lag between idea and practice. With the speed of the new technology, symbolized by the space satellite, this fifty-year wait simply cannot be tolerated. We are living in an era of fantastic change and everyone in education, from the superintendent to the janitor, must recognize this fact and be prepared to act upon it.

Dean Irving R. Melbo, in addressing a conference of school administrators, cited a need at the administrative level for educational statesmen. He then went on to define the characteristics of such a statesman, including the ability to recognize issues. In speaking to this point, he emphasized this modern problem of change in education:

[1] See, for example, numerous speeches and statements by Alvin C. Eurich, Executive Vice-President of the Fund for the Advancement of Education, including his 1956 Statement to the Association for Higher Education (13).

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The ability to recognize issues is all too often lacking among our educational administrators. But perhaps the fault lies with the institutions which trained them, because no man can recognize issues if he has not learned to think in terms of issues.

The simple truth is that great economic and social forces flow with a tidal sweep over communities that are only half conscious of that which is befalling them. Wise statesmen are those who foresee what time is thus bringing, and endeavor to shape institutions and to mold men's thought and purpose in accordance with the change that is silently surrounding them. (23:2)

And, as the Second Industrial Revolution with its automatic controls so sweeps silently into our communities, all schoolmen would do well to remember the words of Whitehead, written over twenty years ago:

Our sociological theories, our political philosophy, our practical maxims of business, our political economy, and our doctrines of education, are derived from an unbroken tradition of great thinkers and of practical examples from the age of Plato . . . to the end of the last century. The whole of this tradition is warped by the vicious assumption that each generation will substantially live amid the conditions governing the lives of its fathers and will transmit those conditions to mold with equal force the lives of its children. We are living in the first period of human history for which this assumption is false. (30:99)

Automation is one of the reasons why the immediate future will bring conditions that cannot be accounted for in the experience of mankind. Automation will radically alter our system of education. It will be well, if we are to recognize and study these issues, to attempt to define this phenomenon.

SOME DEFINITIONS

Automation is one of those concepts capable of several definitions. This is, no doubt, due to the fact that its meaning has not yet been fully developed. At present, automation is a term better described than precisely defined; automation represents a territory that can be mapped, but not surveyed to the last decimal point. Drucker, in his very illuminating discussion, states that "automation can be defined simply, though superficially, as the use of machines to run machines" (10:41).

Macmillan indicates the range of meaning inherent in the term. He speaks of

. . . a great extension in the use of automatic devices, to take over the simpler kinds of work previously done by the men in charge of machines.

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The word "automation" has been applied to this process and widely adopted. It was first used several years ago by engineers of the Ford Motor Company in the United States to describe their methods for automatically conveying workpieces between successive machines; but since then it has come to have a much wider significance, implying any process in which the lower functions of a human operator—both physical and mental—are taken over by self-acting devices. Interpreted in this way, the word "automation" clearly means more than mere mechanization. . . . (22:2)

The CIO Committee on Economic Policy refers to a three-level description developed by Baldwin and Shultz of MIT. The three levels are:

1. The level of *integration* or *"Detroit automation,"* which means the linking together of conventional machines so that the work proceeds from one to the other without human intervention (referred to by Macmillan in the statement about Ford engineers above).

2. The level of "feedback" control devices, which means the introduction of servo-mechanisms or other devices which operate machines without any human control. Adjustments of work output, etc., are governed by an automatic comparison with some standard with the information so derived fed back to the operating device to further govern its work.

3. The level of information control, which means the introduction of computing machines into the system. The computing machines record and store information, make computations and feed the results back into the system, establishing complete control of the whole operation. (6:3-4)

The United Automobile Workers, reflecting the concern of their president, Mr. Walter Reuther, have conducted extensive studies of automation. In a special report to the union, the UAW Education Department referred to automation by its characteristics, stating that automation involved:

1. *Information* that is fed to a
2. *controller*, which operates (and supervises to make sure the operation is done corectly)
3. a *servomechanism* which in turn operates the
4. *machine*, which can be a drill, a milling machine, a lathe, a typewriter, or anything else which serves a purpose, and all this happens in a self-correcting system which is known as a
5. *feedback circuit.* (29:30-31)

EXAMPLES OF AUTOMATION

As will be shown later, the descriptions above represent only the hard core, mechanical attributes of automation, and indicate that the concept, as it is developing today, has abstract, ideational properties which may be even more important. However, these

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abstract properties are not possible without the solid grounding in industrial and scientific developments. To increase understanding, then, of the basis for these descriptive definitions, some examples of automatic production may be helpful.

The following examples, for the most part, are taken from the CIO report (6). Others could be cited. The Cleveland plant turning out Ford engines under the "Detroit automation" system links a battery of machines on a line 1500 feet long with automatic machine tools performing over 500 operations. The plant processes 154 engine blocks an hour with 41 workers. The same operation, under older mass-production methods, would take 117 workers.

The *Wall Street Journal* is the authority for the statement that Raytheon's television and radio division plant will be able to turn out 1000 radios a day with only two workers on the line, as compared with a former need of 200 workers. Office automation (this involves computers and other machines, but also structured planning) is shown in the John Plain mail order house in Chicago, where a computer handles 90,000 tallies a day in keeping track of an 8,000 item catalog. Insurance companies now going to computer operation are represented by one cited which makes 7,000 daily billings, 20,000 monthly calculations for dividends, and 130,000 calculations for agency commissions at a saving of from 175 to 225 clerical personnel. Finally, it is reported that General Mills has produced an automatic machine which will produce electronic equipment. This machine, it is claimed, will require only two workers and a supervisor and has a capacity of over 200,000 assemblies a month.

While these are examples of automation in office and factory, it should be emphasized that they are, as yet, examples. Outside of modern oil refineries, which are almost fully automatic, there are practically no fully automatic plants yet in operation. As John I. Snyder, Jr., president of U. S. Industries, points out, many industries are coming closer to it and have already automatized one or more *parts* of the production process (28). His own U. S. Industries was, in 1955, operating the only fully automatic factory—a government-owned, 155-mm shell plant in Rockford, Illinois. However, a recent study by the Research Institute of America of 1000 major companies found that 16 per cent saw automation as a factor in their industry at the present time and that an additional 23 per cent predicted extensive automation in

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their industry within a decade. Only one-third of the respondents felt that automation would never affect them (4).

CHARACTERISTICS OF AUTOMATION

When automatic operations of various kinds are looked at together, certain characteristics seem to emerge. These characteristics are not mechanical or technical in a machine sense. Rather, they are conceptual, and, as such, have educational significance.

The first, and probably the most important characteristic of the new industrial revolution is the concept of *systems*. While this concept has been discussed before in connection with audio-visual administration, its pervasiveness in business and industry—in any large scale organization—has yet to be realized. The systems idea is best grasped in connection with an office which is undergoing automation. While many processes—billing, posting, etc.—will be done by automatic machinery in such a set-up, the important elements are the planning necessary for the whole procedure, the location and function of necessary personnel, the types of forms used, etc. A system has to be created which gears all of these elements into a smooth flow, with controls feeding back information throughout the process. The system is much more important than the computer which calculates interest rates, for example. A representative of a firm which installs automatic processing in offices was heard to remark recently that it would take his firm eight years to install a system in a large company.

The problem of systems and systems analysis is complex and cannot be treated here at any length. Much automatic machinery can do certain routine work infinitely better and faster than man. Aircraft warning devices are an example of this. But somewhere in any operation, there must be men. Men, machines, and information have all to be coordinated into a system which may begin when a prospect is furnished an insurance salesman or a blip appears upon a radarscope, and ends with the widow receiving a check or the refueling of an interceptor plane after its mission has been accomplished. No longer can the in-between operations be considered piecemeal. Automation means system integrated with system.

Secondly, such systems and such potentials for rapid production need control. The old fairy story of the salt-making machine

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that went on and on grinding salt could easily be translated into a modern engineer's nightmare. There are many elements here, and at different levels. At the level of the mechanics of the production line, information is needed constantly as to the condition of the machines and the work; at the marketing level, information is needed as to style, inventory, price, etc. If a machine gets too hot, information must come back to a control-point and orders issued for lubrication, reduction of speed, or some other remedy; if too many of the wrong color widgets are popping off the end of the line, information again must go to a control-point so that fewer pink widgets and more blue ones are produced. The constant here, and a general characteristic of automatic operation, is the flow of information and the control of the system by that information.

The third general characteristic of automation springs logically from the two which precede it. The systems and control characteristics of automation demand *thorough analysis and long-range planning*. The huge capital involved, for example, in automatizing an operation [2] means that the operation cannot be carried on by caprice. Once an automatic operation is underway, it is very costly to stop it for extensive changes. Hence, accurate planning is required. This planning is, obviously, one of the main factors in the eight-year installation problem described above in connection with another insurance firm.

Planning, however, cannot proceed without analysis. Neither can a system function indefinitely without analysis. This is the reason that the terms "operations analysis" or "operational research" are closely allied to the development of automation, and, for that matter, of all modern technology. The term, operational research, is usually used to refer to the study of management problems from the point of view of the mathematician, the physical and the biological scientist. The operational research staff, for example, in a military headquarters, attempts to strip a problem down to its bare essentials, create a mathematical model of it, and solve for the best answer. As Dr. John Abrams of the Royal Canadian Air Force pointed out (1), in military operational research, use is made of such things as probability statistics, the theory of games, search problems, and the newer theories of value

[2] According to a story in the *Los Angeles Times*, the Farmers Insurance Group in Los Angeles recently installed an IBM 705 computer to control automatic operations. The cost was quoted as a rental of $36,000 per month.

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and decision. These, of course, are all in the realm of mathematics. Methodologies from other sciences are also used.

While there is no doubt that the armed forces of several countries have pioneered in the area of operational research, the term is now being heard in reference to similar problems in industry. The very existence of computers as part of the automation process makes possible the solution of complex mathematical problems relating to operational research in a very short time. For example, in a recent issue of the *Harvard Business Review*, Gaumnitz and Brownless review the mathematics that should be required of managers (19). Decision-making in all forms of automation processes and complex systems requires long-range planning and operational analysis.

The mention of decision-making in connection with automation highlights the fourth characteristic of the process which is *decision-making*. Drucker points out that, in order to control such processes and systems and to make and carry out the necessary plans, many more better educated managers will be needed in the future (10). Decisions dealing with a complex system will have complex consequences. The implications here for education are tremendous.

Finally, although it should be obvious, it is necessary to point out that automation and its clusters of systems is a high order product of technology. That is, it is a *highly technical* business. While the operations themselves may be reduced to watching an occasional gauge or checking the accuracy of a bill for an insurance premium that has just come through a machine, the machines themselves and the systems within which they operate are exceedingly complex. Maintenance, for example, will require very able technicians.[3] The engineering behind some of the equipment involved in automation also presents great problems of design.

These, then, are some of the characteristics of automation and its associated processes and systems: (a) the concept of systems, (b) the flow and control of information, (c) scientific analysis and long-range planning, (d) an increase in the need for wise decision-making, and (e) a high level technology.

[3] For anyone who has not experienced this reality, let him see an ordinary automobile mechanic attempt to solve the problem of making a modern automatic transmission function when it has been jammed or otherwise damaged.

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EFFECTS ON EDUCATION

Those who sense that all social and economic forces affect education in general, as well as specific ways, will realize that a list of the effects of automation on education could be almost endless. Obviously, there are concerns about technological unemployment, even if it may be temporary, and the age structure of our population for the next twenty years will bring less people into the labor force than we need. The need for maximum productivity for prosperity placed against a reduction in the labor force will hurry, rather than stay, the development of automation. A shorter work week seems to be inevitable. Industry will be interested, as never before, in a predictable expanding stable market—the risks are too great in capital, etc., to break down an automated system. This would imply that industry may have a stake in some kind of an annual wage. The initial high costs of capitalization for completely automatic production may operate to further change our economy in the direction of more big business. One authority has suggested that, with a great sameness of product possible from automatic factories, the day of the artisan and hand-worker may once again bloom (25).

All of these and many more possible social and economic effects have educational implications. It is too early to say what many of them will actually turn out to be. However, there are a number of more direct effects that can be identified and these effects will be pronounced at all levels of education. Consider carefully the following quotations:

> Automation's most important impact will not be on employment but on the qualifications and functions of employees.
>
> And this increase both in the numbers of managers and in the demands made on them may well be the largest of all the social impacts of Automation.
>
> One large manufacturing company (now employing 150,000) figures that it will need seven thousand college graduates a year, once it is automated, just to keep going; today it hires three hundred annually.
>
> But the need is above all qualitative—for better educated people. The "trained barbarian," the man who has acquired high gadgeteering skill, will not do. Even in routine jobs, Automation will require ability to think, a trained imagination, and good judgment, plus some skill in logical methods, some mathematical understanding, and some ability well above the elementary level to read and write—in a word, the normal equipment of educated people. Under Automation, a school could do a student no greater disservice than to

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prepare him, as so many do today, for his first job. If there is one thing under Automation that is certain, it is that the job—even the bottom job—will change radically and often. (10:44-45)

In the conditions of modern life, the rule is absolute. The race which does not value trained intelligence is doomed. Not all your heroism, not all your social charm, not all your wit, not all your victories at land or at sea, can move back the finger of fate. Today we maintain ourselves. Tomorrow science will have moved forward yet one more step, and there will be no appeal from the judgment which will then be pronounced on the uneducated. (5)

Information theory is the science underlying all exchange of information, and is of especial importance in the development of telecommunications and automatic control systems. The treatment adopted by the author has enabled him to *give a clear account of recent developments without requiring a knowledge of mathematics in excess of that which may reasonably be expected from the professional engineer or physicist.* (From a book jacket—italics supplied.)

The First Industrial Revolution, the revolution of the "dark satanic mills," was the devaluation of the human arm by the competition of machinery. There is no rate of pay at which a United States pick-and-shovel laborer can live which is low enough to compete with the work of a steam shovel as an excavator. The modern industrial revolution is similarly bound to devalue the human brain, at least in its simpler and more routine decisions. Of course, just as the skilled carpenter, the skilled mechanic, the skilled dressmaker have in some degree survived the First Industrial Revolution, so the skilled scientist and the skilled administrator may survive the Second. *However, taking the Second Revolution as accomplished, the average human being of mediocre attainments or less has nothing to sell that it is worth anyone's money to buy.* (Quoted in 26:13. Italics supplied.)

These quotations are enough to give any thoughtful educator pause. Some of their possible implications and the implications in general of automation may be discussed under four headings: (a) the curriculum and program of studies, (b) administration, (c) the possibilities in automatizing the process of education itself, and (d) changes required in theory. These will be briefly delineated.

The first problem under the program of studies category is the role and placement of subject matter. It can be suggested that those educators who relegate "subject matter" to limbo or who suggest that "we only work through subject matter to something else" are echoing the cliches of the 1920's without meaning in 1957. For certain groups of students, subject matter *per se* will have a new importance. This is not to suggest that all the gains in dealing with individual persons, in attempting to change behavior through

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problem-solving, etc., need be thrown out. It does suggest that a technological society needs technical information in the possession of a rather large group of its members.

Specifically, mathematics will continue to play a greater role than ever in our lives. A thorough revision of the classical mathematics curriculum at the secondary level is called for, in addition to a strengthening of the program at the elementary level. Solid geometry, for example, is according to many modern mathematicians, an interesting but obsolete subject and should be dropped. The high school mathematics program can be revised (and this is being done in a few places) so that, for example, the elements of calculus will be taught to *some* students in high school.

The science phases of the elementary curriculum need revision and strengthening. In many elementary schools in the United States, science instruction is a joke. If so, this may be a pretty high price for our society to pay for a few laughs. Actually, at least for some children, the self-contained classroom in the upper elementary level with its extreme emphasis on social living should be due for a thorough going over. It may be that for some children, we will want some form of departmentalization with a pushing downward of some of the elements of the present high school curriculum. This will mean an earlier transition from a developmental curriculum than most people in elementary education are now willing to make.

Another concept that must be re-examined—and at the very highest levels of education—is the concept of general education. The last twenty years have brought great pressures for general education—for an education so that we may appreciate our common heritage, etc., etc., etc. Survey courses, Great Books, the Harvard essentials, "exploration" courses, and work-experience programs are all examples of this sort. There has been a great deal more worry among the Mark Van Dorens, Mortimer Adlers, and A. Whitney Griswolds about whether or not our technicians, engineers, and scientists could talk to each other than whether or not they were good engineers, scientists, or technicians (I include the social sciences here). The spokesmen for general education have, actually, attempted to create a *specialist in general education* under the guise of dealing with the well-known ignorance of specialists.

It is a moot point whether or not a highly technical society can

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survive and maintain itself without highly specialized technological know-how. It takes time to acquire this know-how. Without decrying general education, it would seem that this extreme emphasis upon it should be re-examined with a view to reducing the time devoted to it. It is recognized that this is in opposition to Drucker's position that the managerial talent of the future will have to have a general education. It is my feeling that Mr. Drucker has become enmeshed in his own net in this instance. Management is, or will become, a highly skilled profession. Mr. Drucker does not want general education; he wants general professional education, and the distinction is important.

The implication of a few paragraphs above—that not all students should have the same education—can be stated in the proposition that education for an age of automation will require homogeneous grouping of students in the public schools and recognition of different patterns of preparation. This is not necessarily to say that an intellectual elite will be created after Grade 4, although this is possible.

The well-known facts of psychology relating to the extreme variations in individuals and the fact that individuals may be homogeneous in one respect and not in others will require a re-examination of even the principle of homogeneous grouping. But it is still reasonable to expect that not all high school students will be able or find it necessary to take the elements of calculus. The great problem here is to find ways and means of shortening and improving the general education periods so that all students can function as citizens, parents, etc., and, at the same time, meet the demands of a technological society for specialists (18). Perhaps the greatest educational dilemma of our time may be stated as follows: *If these two objectives come into complete conflict, a technological society, in order to survive, has no choice but to pick technology.* We may not like the world we so create (and some possibilities in that direction will be discussed in the third paper in this series), but we must recognize it for what it is.

Computers, cyclotrons, and magnetrons—all paraphernalia of the age of automation—point up another great fact that is powerful enough to shake the structure of the whole curriculum from the kindergarten to the graduate school. I refer here to the almost unbelievably rapid growth of knowledge. This is a problem which most curriculum people have ignored by the simple process of duck-

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ing their heads in the sand of "attitude formation" and "problem-solving skills." Incidentally, it has also been completely ignored by their opposite members of the *Saturday Review* set who still think that Aristotle had no need to count the number of teeth in his wife's mouth. The poets, the Greek plays, the socio-drama, and the resource unit do not necessarily have the answer to this problem. The solution must lie in thorough revision of content and the development of means to teach more to more people in less time.

The systems concept and related matters emphasize the fact that, while group dynamics has not yet got beyond the stage of a primitive religion in most curriculum circles, it has the key to many problems that will beset an automatized world. Human relations, group relations, even international relations, become proper subjects of study and necessary areas in which to develop real skill and proper attitudes. Again, here, the problem may be methodological. The solution will not be found in the technique used by a professor of education in holding a "buzz session" or in having committee reports made in class—any more than it will be found in some of the so-called "work-shops" run by school districts under the euphemism of an in-service training program. We need to develop content and method which yields measurable results in this field.

There are many other implications for a curriculum content in an age of automation. To create decision-makers means that the thought processes of students must have experienced conflicts and made decisions; to analyze, to plan, to create systems and to live within these patterns implies an ablity to think in patterns, to take the long-range view, to be creative. Those information-mongers, those fact-purveyors who felt that the position stated above with regard to subject matter was their meat had better keep this fundamental in mind. No mountain of facts will do more than provide the background for this process. Our new curriculum must do so, but not in any sloppy way, based on the assumption that if the children learn how to solve the problem of waste-paper in the cafeteria over a semester's period of time, they have demonstrated outstanding ability in pattern-thinking.

The problems of additional leisure time, the problems of vocational training in a world where vocations will change and buttons or magnetic tapes take their place all need consideration for the new curriculum. The best that we can say is that we have had

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curriculum challenges before, but never of the intensity and urgency that the coming age of automation presents us.

The second category of possible effects is that of administration. As I have discussed some of these problems elsewhere (15), (16), (17), I will merely enumerate them here. The new management concepts—systems, operational research, the use of the theory of games, etc.—all have much to offer school administrators. The theory of school administration—in my opinion, already somewhat creaky—needs a thorough re-examination in the light of these new developments. It is not overstating the case to say that education for an age of automation needs an administrative theory that is at least able to keep up with it. This is a great challenge to theorists in the field of administration and to practicing administrators. If administrators are to lead, it would be well to begin by streamlining their own operations to fit the demands of a new world.

Third, much needs to be done in considering how far we may go in automatizing the process of education itself. Here, as was mentioned before, lies the direct relationship between the audio-visual field and the concepts of automation. This topic will be the subject of the second of these studies. Can the somewhat cold process of closed-circuit television, for example, increase educational efficiency without losing something in the process?

Finally, when all this is added up, what changes can be expected in our intellectual structure—in the content of "education" itself? Certain things have been mentioned—a re-examination of curriculum theory, especially as it applies to the program of studies, and of administrative theory. It is to be expected that new bodies of such theory will be created and studies conducted on hypotheses based upon these theories.

Actually, education, as a subject of study, has stood on a plateau for about 20 years. The audio-visual field is one exception to this statement. The curriculum books still say much the same thing; the speakers at institutes and meetings are still re-working the old material. Education, as a discipline, is in the position it was around 1900 when a new graduate content was created almost overnight by Thorndike, Dewey, Monroe, and a few others.

We are on the edge of a breakthrough. It will, in fact, be forced upon us by circumstances if we do not make it ourselves. A new content for education as a discipline needs to be made. Already, beginnings can be detected. For example, most of our

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psychology—and certainly all of it that reaches the teacher-training level—is individual learning psychology with overtones of growth and development. Social psychology—the study of the behavior of people in groups, their responses to mass communications, etc.—has emerged as a separate discipline. Much can be gained by working pertinent material from this important field into teacher education programs (again, the audio-visual field has tapped social psychology more for its data than any field in education.)

New philosophies of education need to be created. The recent NSSE Yearbook is rich in suggestions in this matter (24). All of the remaining social sciences—anthropology, political science, international relations, etc.—can contribute much. Experiments in the biology of the mentally retarded, new advances in psychiatry, and the new stress theory of disease can all give a little to the new content in education. We need now to work with experts in these disciplines and get the material flowing into our thinning lifeblood that still carries the weak corpuscles of "life-adjustment" and "purposive activity." The age of automation will require it.

One final word as to the content of education as a discipline. Much has been heard in recent years from critics of education, such as Arthur Bestor, that "there ain't no such thing" as a content in education. Bestor's position, of course, is untenable. However, it has, no doubt, been true that some courses in education have been sadly lacking in the content that was there for the asking. The new content that the age of automation suggests will be much more demanding intellectually than that we already have. The problems of teacher education will become, therefore, more acute. No true profession in the age of automation will be able to slough off its intellectual obligations. This includes teachers now who consider a course difficult when it has two twenty-minute examinations.

SUMMARY

An attempt has been made in this paper to describe automation, both as an industrial process involving flow and control and as an intellectual system involving planning and decision-making. General effects on education in the areas of curriculum and program of studies, administration, general education, organization for instruction, the necessity for teaching new patterns of think-

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ing, vocational and technical education were discussed. The concept of the possibility of automatizing the instructional process itself was introduced, and the outlines of a new content for education as a discipline were suggested. The following paper will deal with the problem of automatizing instruction through audio-visual means, and the third and last paper in the series will consider the dangers and difficulties involved in such a process.

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